

Concrete-Domain Closure under AASC

A Non-Dischargeability Theorem for Bounded Realized Persistence

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Abstract

This article establishes a concrete-domain closure theorem under AASC. It does not assume concrete reality as a primitive premise and does not identify concreteness with a particular physics, spacetime model, material ontology, observer structure, biological process, or semantic theory. The target is the fixed-domain closure of realized persistence, repair-localized standing, bounded consequence retention, trajectory irreversibility, role-anchor fixation, quotient stability, and fail-closed continuation.

The central claim is non-dischargeability. Realized persistence contains irreversible standing-bearing consequence retention; repair localization contains bounded lawful correction of standing-relevant degradation without identity reset, illicit same-act repair, or unrestricted global reindexing; bounded standing contains a fixed repair-localized standing locus. Abstract-only, formal-only, trace-only, semantic-only, skin-only, global, diffuse, AMetric, and bookkeeping closures each remove or evade one of those fixed-domain obligations. Every non-concrete closure therefore factorizes through loss of realized persistence, loss of repair-localized standing, loss of bounded consequence retention, semantic discharge without consequence retention, formal encoding without participation, AMetric non-selection, skin-only presentation, scope change, or inert bookkeeping.

The main theorem is

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow \neg \text{NonConcClosure}_D.$$

Consequently, concrete-domain participation is obtained as the admissible same-domain closure class:

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow [\mathcal{C}_D]_{\equiv \text{Conc}}.$$

Here $[\mathcal{C}_D]_{\equiv \text{Conc}}$ is the concrete-domain participation equivalence class: the admissibility-bearing closure class in which realized persistence and bounded repair-local standing participate in the domain rather than being discharged into description, trace succession, semantic reference, representation, or formal bookkeeping. Concrete-domain participation is not stipulated; it is the admissible closure forced because realized repair-localized bounded persistence is non-dischargeable into abstract-only, formal-only, trace-only, semantic-only, representational, global, diffuse, AMetric, or bookkeeping structure.

Theorem overview. Realized, repair-localized, bounded standing-bearing persistence is non-dischargeable into abstraction, formal inscription, trace succession, semantic reference, skin presentation, global or diffuse standing, AMetricity, or bookkeeping; it induces concrete-domain participation up to admissibility-preserving equivalence.

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1 Scope and Target Discipline

This article proves concrete-domain closure in the AASC sense. It does not prove a particular physical theory, a particular universe, a preferred spacetime model, ordinary matter, biological life, observers, consciousness, or a substance ontology. Its target is narrower and stronger: the admissible closure of realized, repair-localized, bounded standing-bearing persistence. In this manuscript, *concrete* means non-dischargeable standing-bearing domain participation; it does not by itself specify any particular physical ontology.

The theorem is an exhaustion and non-dischargeability theorem. It does not begin by asserting concrete reality. It begins with a target class fixed by established AASC structure:

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D .$$

It then shows that this target is non-dischargeable into abstraction, formal notation, trace succession, semantic reference, skin presentation, global standing, diffuse standing, AMetric pre-admissibility, scope change, hidden selector import, or inert bookkeeping. Concrete-domain participation is the fixed-domain closure that remains after those non-concrete discharge routes fail.

1.1 Licensed readings

The following readings are licensed:

- (a) a fixed-domain closure theorem from bounded realized persistence to concrete-domain participation;
- (b) a non-dischargeability closure result for realized, repair-localized, bounded standing-bearing persistence;
- (c) an exhaustion theorem against abstract-only, formal-only, trace-only, semantic-only, skin-only, global, diffuse, AMetric, and bookkeeping closures;
- (d) a theorem that concrete-domain participation is forced up to admissibility-preserving equivalence, not by representative selection;
- (e) the framework-internal claim that reality is concretely participated rather than merely described.

1.2 Forbidden readings

The following readings are not licensed:

- (a) a derivation of a specific physical model, spacetime theory, particle ontology, cosmology, or empirical law;
- (b) a proof that matter, observers, consciousness, organisms, or biological life are forced by this theorem alone;
- (c) a claim that a canonical concrete representative is selected;
- (d) a claim that a formal description is thereby identical with its concrete-domain participation;
- (e) a claim that semantic reference, observation, or representation is the source of concreteness;
- (f) a claim that every formal system is concrete merely because it is describable.

2 Structural Dependencies for Concrete-Domain Closure

The proof uses established AASC closure and induction results as structural dependencies. They are not empirical postulates and do not smuggle a physical ontology into the theorem.

D1. **Null closure.** Total AASC vacuity cannot occupy the admitted side of fixed-domain admissibility-bearing evaluability.

D2. **Governance induction.** Non-degenerate continuation induces a non-vacuous governance equivalence class:

$$\text{NonDeg}(D) \rightarrow [G_D]_{\equiv_{\text{Gov}}}.$$

D3. **Continuation-locus induction.** Non-degenerate standing-bearing continuation induces a continuation-locus equivalence class:

$$\text{NonDegCont}(D) \rightarrow [L_D]_{\equiv_{\text{Loc}}}.$$

D4. **Role-anchor fixation.** Non-selective continuation loci under admissible boundary-trace fixation induce a role-anchor equivalence class:

$$\text{NonDegCont}(D) \wedge \text{BoundaryTrace}_D \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

D5. **Persistence-regime realization.** Irreversible standing-bearing continuation induces a realized persistence-regime equivalence class:

$$\text{IrrevStandCont}(D) \rightarrow [\mathcal{P}_D]_{\equiv_{\text{Pers}}}.$$

D6. **Standing localization.** Realized persistence with lawful repair semantics induces bounded standing-localization equivalence:

$$\text{RealPers}_D \wedge \text{RepairSem}_D \rightarrow [\mathcal{B}_D]_{\equiv_{\text{Bound}}}.$$

D7. **AMetric boundary discipline.** A pre-admissible boundary supplies no selector, metric, ranking, default, probability, canonical representative, or transport rule unless admitted structure is added.

D8. **Anchor–tensor–skin decomposition.** Standing-bearing authority is anchor or tensor load; skin changes presentation but carries no gate, repair, persistence, or standing load.

- D9. **Bivalence and fail-closed repair.** There is no third same-domain status such as non-concrete-but-standing, abstract-but-realized, trace-only-but-persistent, or skin-only-but-authoritative.
- D10. **Fixed-domain exhaustion.** Same-scope closure must preserve identity, transport, quotient, standing, boundary, repair, and continuation structure; otherwise it is scope change, bookkeeping, AMetric inertness, or inadmissible import.

The present closure result is:

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow [\mathcal{C}_D]_{\equiv_{\text{Conc}}}.$$

3 Definitions and Type Discipline

Definition 3.1 (Concrete-domain participation). A concrete-domain participation class $[\mathcal{C}_D]_{\equiv_{\text{Conc}}}$ is the equivalence class of admissibility-bearing domain participation structures in which realized persistence possesses irreversible, repair-localized, bounded standing-relevant consequence retention that cannot be discharged into purely representational, formal, semantic, trace-level, skin-level, global, diffuse, AMetric, or bookkeeping structure without destroying the realized persistence target. Equivalently, it is the positive participation structure preserving realized persistence, repair-localized standing, bounded consequence retention, trajectory irreversibility, role-anchor fixation, boundary-trace compatibility, quotient stability, and fail-closed standing failure under lawful same-domain admissibility-preserving redescription.

Concrete-domain participation is therefore defined positively by standing-bearing domain participation, not merely by negation. It is not a selected object, observer, physical body, material substance, spacetime point, or canonical representative. It is the admissibility-bearing participation class whose removal discharges the target into a mode that cannot preserve realized repair-localized bounded standing.

Definition 3.2 (Concrete-domain equivalence). Two concrete-domain participation structures $\mathcal{C}$ and $\mathcal{C}'$ are concrete-equivalent, written $\mathcal{C} \equiv_{\text{Conc}} \mathcal{C}'$, iff they preserve the same realized persistence, repair-localized standing, bounded consequence-retention, trajectory irreversibility, role-anchor fixation, quotient behavior, and fail-closed standing behavior under lawful same-domain admissibility-preserving redescription.

Concrete-domain equivalence is always modulo lawful same-domain admissibility-preserving redescription. It is not syntactic sameness and does not select a privileged representative.

Definition 3.3 (Non-concrete closure). NonConcClosure_D holds iff a candidate closure of $\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D$ is made through one or more of the following: abstract-only description, formal-only inscription, trace-only succession, semantic-only reference, skin-only presentation, global standing, diffuse standing, AMetric pre-admissibility, hidden selector import, scope change, or inert bookkeeping.

Definition 3.4 (Abstract-only closure). AbstractOnly_D holds iff the candidate closure is a structural description without standing-bearing domain participation preserving realized persistence, repair-local standing, and bounded consequence retention.

Definition 3.5 (Formal-only closure). FormalOnly_D holds iff the candidate closure is given by formal inscription, syntax, symbol manipulation, or proof record without realized standing-bearing domain participation.

Definition 3.6 (Trace-only closure). TraceOnly_D holds iff the candidate closure supplies succession, update, or trajectory trace without irreversible standing-bearing consequence retention.

Definition 3.7 (Semantic-only closure). SemanticOnly_D holds iff the candidate closure is reference, meaning, denotation, representation, or interpretation without repair-localized standing-bearing domain participation.

Definition 3.8 (Skin-only closure). SkinOnly_D holds iff the candidate closure changes presentation while supplying no anchor or tensor load for realized persistence, repair localization, bounded standing, or fail-closed continuation.

Definition 3.9 (Concrete closure target). The concrete closure target is

$$\Omega_D := \text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D.$$

The theorem concerns only domains in which Ω_D is in force.

4 Realized Persistence and Repair-Localized Standing

Concrete-domain closure begins only after realized persistence and bounded repair-localized standing are already in force. This prevents two overclaims. First, the theorem does not infer concreteness from mere formal description. Second, it does not infer concreteness from mere continuation or from realized persistence alone. The target includes irreversible standing-bearing consequence retention, bounded repair-localization, and non-dischargeable domain participation.

Proposition 4.1 (Target commitments). *If Ω_D holds, then the following are fixed-domain commitments:*

$$\text{RealPers}_D, \quad \text{RepairLoc}_D, \quad \text{BoundStand}_D, \quad \text{Irrev}_D, \quad \text{ConseqRet}_D, \quad \text{FailClosed}_D.$$

Proof. RealPers_D is realized persistence and therefore includes irreversible standing-bearing consequence retention. RepairLoc_D is repair-localization and therefore includes bounded lawful correction of standing-relevant degradation without identity reset, same-act repair, or global reindexing. BoundStand_D is bounded standing-localization and therefore fixes a bounded locus for standing-bearing consequence retention. Fail-closed handling is inherited from AASC standing failure non-repairability. These are not optional supplements; they are the typed components of Ω_D . $\square$

Remark 4.2 (Concrete is target-relative). The word concrete in this article is target-relative. It means admissibility-bearing domain participation preserving Ω_D . It does not mean a primitive substance ontology or a particular physical theory.

Theorem 4.3 (Realization alone is not concrete-domain closure).

$$[\mathcal{P}_D]_{\equiv \text{Pers}} \not\Rightarrow [\mathcal{C}_D]_{\equiv \text{Conc}}$$

unless repair-localized bounded standing structure is also in force.

Proof. A realized persistence-regime equivalence class preserves irreversible standing-bearing trajectory participation. Concrete-domain participation additionally requires repair-localized bounded consequence retention that cannot be discharged into formal, semantic, trace-level, skin-level, global, diffuse, AMetric, or bookkeeping structure. Without RepairLoc_D and BoundStand_D , realized persistence may remain a realized persistence regime without closing the non-concrete routes addressed in this article. Thus realization is a dependency of concrete-domain closure, not its synonym. $\square$

Proposition 4.4 (Non-dischargeability target). *The concrete-domain closure theorem is equivalently a non-dischargeability theorem for Ω_D :*

$$\Omega_D \rightarrow \text{NonDischarge}_D(\text{AbstractOnly}, \text{FormalOnly}, \text{TraceOnly}, \text{SemanticOnly}, \text{SkinOnly}, \text{GlobalStand}, \text{DiffuseStand})$$

It shows that the fixed target cannot be preserved after discharge into any non-concrete closure mode.

5 Abstract-Only Closure Failure

The first non-concrete route says that bounded realized persistence may close as pure abstraction.

Theorem 5.1 (Abstract-only closure failure).

$$\Omega_D \wedge \text{AbstractOnly}_D \rightarrow \perp.$$

Proof. Assume Ω_D . Then realized persistence, repair-localized standing, bounded consequence retention, and fail-closed continuation are in force. Abstract-only closure removes standing-bearing domain participation while retaining only structural description. But repair-localized standing requires a bounded locus over which degradation, correction, failure, identity retention, and retained consequence are evaluated. A purely abstract closure supplies no such participating locus. If the abstraction changes no standing-bearing use, it is bookkeeping. If it preserves the use, it is no longer abstract-only: it carries anchor or tensor load in the concrete-domain participation structure. Hence Ω_D and AbstractOnly_D cannot jointly hold. $\square$

Corollary 5.2 (Description is not participation). *A description of realized bounded persistence is not, merely as description, the domain participation by which that persistence is realized.*

6 Formal-Only Closure Failure

The second route says that formal inscription, syntax, or proof-record availability is enough for closure.

Theorem 6.1 (Formal-only closure failure).

$$\Omega_D \wedge \text{FormalOnly}_D \rightarrow \perp.$$

Proof. Formal structure may describe, encode, or reason about Ω_D . But Ω_D includes irreversible standing-bearing consequence retention and repair-localized bounded standing. Formal-only closure supplies a record or expression while withholding the standing-bearing domain participation through which degradation, correction, identity retention, and fail-closed consequence are carried. If the formal structure itself performs that standing-bearing repair-local participation, it is no longer formal-only. If it does not, it is a description without realized persistence participation. Therefore formal-only closure cannot close the target. $\square$

Theorem 6.2 (Formal encoding insufficiency).

$$\text{FormalOnly}_D \rightarrow \neg \text{RealPers}_D$$

whenever formal encoding is not accompanied by irreversible repair-localized standing-bearing participation.

Proof. A formal system may encode relations corresponding to persistence, repair, trajectory, or bounded standing. Encoding is not identical with standing-bearing participation. If the formal structure preserves irreversible repair-localized consequence retention as part of the domain, it has crossed into anchor or tensor load and is no longer formal-only. If it does not, it remains an encoding without realized participation. Hence formal-only closure cannot preserve RealPers_D for the concrete closure target. $\square$

Remark 6.3 (Formalism boundary). This theorem does not deny the admissibility of formal systems. It denies that formal inscription alone closes realized, repair-localized, bounded standing-bearing persistence.

7 Trace-Only Closure Failure

The third route says that succession, update trace, or trajectory logging is enough for closure.

Theorem 7.1 (Trace-only closure failure).

$$\Omega_D \wedge \text{TraceOnly}_D \rightarrow \perp.$$

Proof. Realized persistence includes irreversible standing-bearing consequence retention. A trace-only closure records succession without retaining standing-bearing consequence under fail-closed continuation. It may exhibit order, update, or event-like sequence, but without standing-bearing irreversibility and repair-localized bounded consequence it is reindexable trace rather than realized persistence. If trace carries the retained standing consequence, it has become anchor or tensor load and is not trace-only. Hence trace-only closure fails. $\square$

Corollary 7.2 (Reindexing collapse). *A trace that can be globally reindexed without standing-bearing loss does not preserve Ω_D . A trace that cannot be so reindexed carries non-trace persistence load.*

8 Semantic-Only Closure Failure

The fourth route says that reference, meaning, denotation, interpretation, or semantic grounding supplies closure.

Theorem 8.1 (Semantic-only closure failure).

$$\Omega_D \wedge \text{SemanticOnly}_D \rightarrow \perp.$$

Proof. Semantic reference may identify, describe, or denote a persistence structure. But the target requires repair-localized bounded standing participation. Reference without such participation does not preserve degradation, correction, failure, bounded identity, or retained standing consequence. If semantic structure determines those standing-bearing features, it performs anchor or tensor work in the domain and is no longer semantic-only. If it does not, it is insufficient to close Ω_D . Therefore semantic-only closure fails. $\square$

Theorem 8.2 (Semantic discharge insufficiency).

$$\text{SemanticOnly}_D \rightarrow \neg \text{StableConseqRet}_D$$

for the concrete closure target.

Proof. Semantic reference can specify, denote, or describe standing-bearing persistence. It does not, as semantic reference alone, preserve irreversible repair-localized consequence retention. If semantic structure carries such retention, it functions as anchor or tensor domain structure and is no longer semantic-only. If it remains semantic-only, it cannot preserve stable consequence retention under repair-localized bounded standing. Therefore semantic discharge is insufficient for concrete-domain closure. $\square$

Remark 8.3 (No semantic grounding interpretation). Concrete-domain closure concerns admissibility-bearing domain participation, not semantic grounding, intentional reference, representational meaning, observational identification, or phenomenological givenness.

9 Skin-Only Closure Failure under ATS

The fifth route says that presentation, representation, display, or surface form is enough for closure.

Theorem 9.1 (Skin-only closure failure).

$$\Omega_D \wedge \text{SkinOnly}_D \rightarrow \perp.$$

Proof. By ATS, admissibility-bearing descriptive content is anchor, tensor, or skin. Skin changes presentation but carries no gate authority, repair-localization load, standing-bearing consequence, or persistence-retention load. The target Ω_D requires realized persistence and bounded repair-local standing. If a presentation preserves that load, it is anchor or tensor content and is not skin-only. If it remains skin-only, it cannot close the target. Hence skin-only closure fails. $\square$

Theorem 9.2 (Concrete closure load).

$$[\mathcal{C}_D]_{\equiv_{\text{Conc}}} \rightarrow \text{Anchor}(\mathcal{C}_D) \vee \text{Tensor}(\mathcal{C}_D), \quad \text{SkinOnly}_D \rightarrow \neg \text{ConcreteClosure}_D.$$

Proof. Concrete-domain participation preserves realized persistence, repair localization, bounded standing, and fail-closed consequence retention. Each is standing-bearing load. By ATS exhaustion, standing-bearing load is anchor or tensor content. Skin-only presentation has no gate, repair, persistence, standing, or consequence-retention authority. Hence concrete closure is anchor/tensor closure, and skin-only closure cannot be concrete closure. $\square$

10 Global and Diffuse Closure Failure

The sixth route says that realized bounded persistence may close through global or diffuse standing rather than concrete-domain participation.

Theorem 10.1 (Global closure failure).

$$\Omega_D \wedge \text{GlobalStand}_D \rightarrow \perp.$$

Proof. Bounded repair-localized standing requires a repair-local locus over which degradation, correction, failure, and identity retention factorize. Global standing refuses that bounded factorization. If global standing is unconstrained, it cannot preserve repair locality under perturbation. If it supplies a hidden global coordinator, that coordinator is non-minimal selector or anchor load and

violates the claimed global closure. If it preserves bounded repair locality, it is not global standing. Hence global closure is incompatible with Ω_D . $\square$

Theorem 10.2 (Diffuse closure failure).

$$\Omega_D \wedge \text{DiffuseStand}_D \rightarrow \perp.$$

Proof. Diffuse standing lacks stable bounded consequence retention. Under perturbation, either standing-relevant consequence is lost, or a hidden coordination structure is added to distinguish preserved standing from drift. Loss contradicts realized persistence. Hidden coordination supplies non-diffuse anchor or tensor load. Therefore diffuse closure cannot preserve Ω_D . $\square$

11 AMetric Boundary and No Default Concrete Selection

The seventh route says that concrete closure may be selected by default, by arbitrary choice, by canonical representative, or by pre-admissible asymmetry.

Theorem 11.1 (AMetric non-selection of concrete closure).

$$\text{AMetricClosure}_D \rightarrow \neg \text{Default}([\mathcal{C}_D]_{\equiv_{\text{Conc}}})$$

unless additional admitted selector structure is supplied.

Proof. AMetric boundary discipline forbids admissibility-bearing selection by default, metric, ranking, probability, parameter, or canonical representative without admitted witness structure. A default concrete representative is therefore unavailable. If a selector is supplied, the closure is not AMetric; it contains admitted non-null structure. If no selector is supplied, no default concrete representative is chosen. Concrete-domain closure is equivalence-class participation, not AMetric representative selection. $\square$

Theorem 11.2 (AMetric non-concretion).

$$\text{AMetricClosure}_D \rightarrow \neg \text{ConcreteClosure}_D$$

unless admissibility-bearing boundary-fixed persistence structure is supplied.

Proof. AMetric pre-admissibility supplies no standing-bearing transport, repair-local consequence retention, role-anchor fixation, boundary trace, quotient stability, or fail-closed persistence. Concrete closure requires all of these as participation structure. If the required boundary-fixed persistence structure is supplied, the regime is no longer AMetric-only. If it is not supplied, no concrete-domain participation is fixed. Thus AMetric closure cannot by itself concretize the target. $\square$

Corollary 11.3 (No canonical concrete representative).

$$[\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\rightarrow \exists! C.$$

The theorem induces a concrete-domain participation equivalence class, not a unique representative.

12 Non-Concrete Closure Exhaustion

This section records the route exhaustion that turns the preceding local failures into a closure theorem.

Theorem 12.1 (Non-concrete factorization exhaustion). *Every same-domain candidate non-concrete closure of Ω_D factorizes through one or more of the following closure modes:*

$$\begin{aligned} \text{NonConcClosure}_D \rightarrow & \text{AbstractOnly}_D \vee \text{FormalOnly}_D \vee \text{TraceOnly}_D \vee \text{SemanticOnly}_D \vee \text{SkinOnly}_D \\ & \vee \text{GlobalStand}_D \vee \text{DiffuseStand}_D \vee \text{AMetricClosure}_D \vee \text{ScopeChange}_D \vee \text{Bookkeeping}_D. \end{aligned}$$

Proof. A same-domain closure must either preserve realized persistence, repair-local standing, bounded consequence retention, role-anchor fixation, trajectory irreversibility, quotient stability, and fail-closed continuation, or fail to do so. If it preserves them as domain participation, it is concrete-domain participation in the AASC sense. If it does not preserve them, the non-concrete attempt factorizes through abstraction, formal encoding, trace succession, semantic discharge, skin presentation, global standing, diffuse standing, AMetric pre-admissibility, scope change, or bookkeeping. Any remaining route either changes the fixed domain, introduces hidden selector structure, becomes inert bookkeeping, or loses realized persistence. The fixed-domain exhaustion theorem rules out additional same-type routes. $\square$

Theorem 12.2 (Non-abstract closure theorem).

$$\Omega_D \rightarrow \neg \text{NonConcClosure}_D.$$

Proof. By the preceding failure theorems, each non-concrete closure factor contradicts realized persistence, repair localization, bounded standing, AMetric discipline, ATS, or same-domain preservation. Hidden selector import is not a non-concrete closure but an admitted non-minimal selector load; scope change abandons the fixed target; bookkeeping changes no licensed use. Therefore Ω_D cannot be discharged into non-concrete closure. $\square$

13 Non-Abstract Closure and Concrete-Domain Participation

Theorem 13.1 (Non-abstract concrete-domain closure). *Let D be a fixed AASC domain. If realized persistence, repair-localized standing, and bounded standing structure are in force, then the target is non-dischargeable into non-concrete closure modes:*

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow \neg \text{NonConcClosure}_D.$$

Proof. Assume $\Omega_D = \text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D$. By non-concrete factorization exhaustion, any candidate non-concrete closure is abstract-only, formal-only, trace-only, semantic-only, skin-only, global, diffuse, AMetric, scope-changing, or bookkeeping. Abstract-only closure lacks standing-bearing domain participation. Formal-only closure describes without realizing participation. Trace-only closure lacks irreversible consequence retention. Semantic-only closure lacks repair-local standing participation. Skin-only closure lacks anchor/tensor load. Global and diffuse closures fail bounded repair-local factorization. AMetric closure supplies no selector, transport, repair, or standing-bearing participation. Scope change fails the fixed-domain target. Bookkeeping changes no licensed use.

Each case contradicts Ω_D or ceases to be a same-domain closure of it. Therefore Ω_D is not dischargeable into non-concrete closure. $\square$

Corollary 13.2 (Concrete-domain participation). *Under the same hypotheses, the admissible same-domain closure is concrete-domain participation up to lawful same-domain admissibility-preserving equivalence:*

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow [\mathcal{C}_D]_{\equiv_{\text{Conc}}}.$$

Proof. Concrete-domain participation is the positive closure class preserving realized persistence, repair-localized standing, bounded consequence retention, role-anchor fixation, trajectory irreversibility, quotient stability, and fail-closed continuation as domain participation. Since non-concrete closure is exhausted and excluded by the preceding theorem, the admissible same-domain closure is the concrete-domain participation equivalence class. $\square$

Corollary 13.3 (Concrete-reality formulation). *In AASC terms, realized repair-localized bounded standing-bearing persistence cannot fail into non-concrete closure. Concrete-domain participation is forced as the admissible closure of the target.*

Remark 13.4 (Concrete-reality reading). The corollary is the fixed-domain AASC form of the claim that concrete reality exists. It does not identify the concrete-domain class with a particular physical model, object inventory, observer, or universe. It says that the realized bounded persistence target cannot be merely abstract, formal, trace-only, semantic, representational, global, diffuse, or AMetric.

14 Concrete Closure Normal Form

Theorem 14.1 (Concrete closure normal form). *Every admissible closure of realized, repair-localized, bounded standing-bearing persistence factorizes through the following equivalence structures:*

$$\text{Closure}_D \Rightarrow ([G_D]_{\equiv_{\text{Gov}}}, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}, [\mathcal{P}_D]_{\equiv_{\text{Pers}}}, [\mathcal{B}_D]_{\equiv_{\text{Bound}}}, [\mathcal{C}_D]_{\equiv_{\text{Conc}}}).$$

Proof. Admissible closure of the target requires governance of licensed standing-bearing use, continuation loci for persistence, role-anchor fixation for locus participation, realized persistence for irreversible consequence retention, bounded standing-localization for repair-local standing, and concrete-domain participation for non-abstract closure. Removing any component sends the closure into one of the rejected non-concrete modes: governanceless continuation, locusless succession, fixationless role drift, trace-only trajectory, global or diffuse standing, abstract-only description, formal-only inscription, semantic-only reference, skin-only presentation, AMetric non-selection, scope change, or bookkeeping. Therefore admissible closure factorizes through the listed structures. $\square$

Corollary 14.2 (Uniqueness up to closure equivalence). *The induced concrete-domain participation is unique up to $\equiv_{\text{Conc}}$, not unique by representative selection:*

$$\text{Closure}_D(\text{RealPers}, \text{RepairLoc}, \text{BoundStand}) \rightarrow [\mathcal{C}_D]_{\equiv_{\text{Conc}}}.$$

15 Finite Witness Model

The finite witness is not a physical model. It is a minimal structural model showing why non-concrete closures fail once realized persistence, repair locality, and bounded standing are fixed.

Let

$$M = \{s_0, s_1, s'_1\}$$

with a realized persistence regime $[\mathcal{P}_M]_{\equiv \text{Pers}}$, bounded standing-localization $[\mathcal{B}_M]_{\equiv \text{Bound}}$, bounded consequence-retention relation K , perturbation $p : s_1 \mapsto s'_1$, repair map $r : s'_1 \mapsto s_1$, irreversible trajectory $t : s_0 \rightarrow s_1$, quotient relation $\sim$, and fail-closed rule F .

Proposition 15.1 (Finite concrete-closure witness). *If M preserves realized persistence, repair-localized standing, and bounded standing under p , r , t , $\sim$, and F , then M induces a concrete-domain participation class:*

$$\text{RealPers}(M) \wedge \text{RepairLoc}(M) \wedge \text{BoundStand}(M) \rightarrow [\mathcal{C}_M]_{\equiv \text{Conc}}.$$

Proof. If concrete-domain participation is removed, the model can only retain a description, trace, semantic reference, representation, or bookkeeping record of s_0, s_1, s'_1 . That record does not preserve standing-bearing consequence through perturbation and lawful repair. If repair localization is removed, standing cannot distinguish lawful correction from replacement or global reindexing. If bounded standing is removed, perturbation cannot be evaluated over a fixed standing locus. If bounded consequence retention is removed, concrete participation collapses into formal trace succession. If irreversibility is removed, the trajectory becomes reindexable trace. If anchor/tensor load is removed, the model becomes skin-only. If boundary fixation is removed, the role-anchor of the persistence locus is unfixed. If quotient stability is removed, presentation drift becomes hidden standing load. If fail-closed handling is removed, failed standing is silently repaired. Therefore every non-concrete removal destroys the target structure or imports hidden load. The finite model induces $[\mathcal{C}_M]_{\equiv \text{Conc}}$. $\square$

Removed or substituted closure	Collapse result
Concrete-domain participation	persistence becomes abstract-only, formal-only, semantic-only, trace-only, or bookkeeping
Repair localization	standing cannot preserve bounded consequence under perturbation
Bounded standing	global or diffuse closure fails repair-local factorization
Bounded consequence retention	concrete participation collapses into formal trace succession
Irreversibility	trajectory becomes reindexable trace
Anchor/tensor load	closure becomes skin-only presentation
Boundary trace	role-anchor fixation is unavailable
Quotient	presentation drift becomes hidden standing load
Fail-closed rule	standing failure is silently repaired
AMetric discipline	default concrete representative selection is smuggled in

The finite witness demonstrates the minimal closure pattern: if realized persistence and bounded repair-local standing are retained, non-concrete closure modes fail locally even in the smallest structural setting.

16 No Specific-Physics Overclaim

Concrete-domain participation is not a specific physics claim. It is a closure class for realized bounded persistence. The following implications are not licensed:

$$[\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \text{Spacetime}_D, \quad [\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \text{Matter}_D, \quad [\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \text{Observer}_D, \quad [\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \text{Life}_D, \quad [\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \text{U}$$

The theorem says that the target participates concretely in the domain. It does not say which empirical ontology, physical law, metric, manifold, field content, organismic structure, or observational structure realizes that participation.

Proposition 16.1 (Concrete participation is not representative realism).

$$[\mathcal{C}_D]_{\equiv_{\text{Conc}}} \not\Rightarrow \exists! C.$$

Proof. Concrete-domain closure is stated at the level of lawful redescription-preserving equivalence. A unique representative requires additional admissible selector structure. A Metric discipline blocks such selection by default. Therefore the theorem induces equivalence-class participation, not representative realism. $\square$

17 Objections and Replies

Objection	Reply
The theorem merely asserts reality	It proves closure by exhausting non-concrete routes for a fixed target.
Concrete is undefined	Concrete means irreversible repair-localized bounded standing-bearing domain participation not dischargeable into abstraction, syntax, trace, semantics, skin, global standing, diffuse standing, AMetricity, or bookkeeping without destroying realized persistence.
Formal structure is enough	Formal structure can describe the target but cannot supply realized repair-local standing participation unless it becomes non-formal load.
Semantic reference is enough	Reference does not carry repair-local consequence retention unless it becomes anchor/tensor structure in the domain.
Trace succession is enough	Trace-only succession lacks irreversible standing-bearing consequence retention.
Skin presentation is enough	Skin has no gate, repair, persistence, or standing load.
Global standing is enough	Global standing cannot factorize bounded repair-local correction under perturbation.
Diffuse standing is enough	Diffuse standing cannot preserve stable consequence retention without hidden coordination.
Concrete closure selects a representative	The theorem induces an equivalence class, not a canonical representative.
The theorem proves a specific universe	It proves concrete-domain participation, not a physical model, cosmology, or object inventory.
Observers are smuggled in	Boundary and trace roles are structural continuation conditions, not observational subjects.
Embodiment is assumed	Bounded standing-localization is weaker than biological or material embodiment.

18 Closure Summary

Closure route	Proposed move	Closure mechanism
Vacuity	terminate in nullity	null-regime closure
Governanceless continuation	continue without constraints	governance induction
Locusless persistence	continue without persistence locus	continuation-locus induction
Fixationless locus	avoid role-anchor fixation	boundary-trace fixation
Trace-only realization	substitute trajectory trace	irreversibility and realized persistence
Global or diffuse standing	avoid bounded standing locus	standing localization and repair locality
Abstract-only closure	describe without participation	abstract-only closure failure
Formal-only closure	inscribe without participation	formal-only closure failure
Semantic-only closure	refer without participation	semantic-only closure failure
Skin-only closure	present without load	ATS closure
AMetric closure	select by default	AMetric non-selection
Scope change	alter the fixed domain	fixed-domain exhaustion
Bookkeeping	change no licensed use	inertness

19 Canonical Results

19.1 Canonical theorem identity

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow \neg \text{NonConcClosure}_D.$$

19.2 Concrete-domain participation corollary

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow [\mathcal{C}_D]_{\equiv_{\text{Conc}}}.$$

19.3 Non-dischargeability formulation

Realized repair-localized bounded standing-bearing persistence is not dischargeable into abstract-only, formal-only, semantic-only, trace-only, skin-only, global, diffuse, AMetric, scope-changing, or bookkeeping closure while preserving the target.

19.4 Plain-language theorem

Realized, repair-localized, bounded standing-bearing persistence cannot close as merely abstract, formal, trace-only, semantic, representational, global, diffuse, AMetric, scope-changing, or bookkeeping structure.

19.5 Concrete-reality formulation

Concrete reality is forced here in the AASC sense of concrete-domain participation: the admissible closure of realized repair-localized bounded persistence after non-concrete discharge routes fail.

19.6 Direct formulation

If standing-bearing persistence is realized, irreversible, repair-localized, and bounded, then it is not merely abstract. It participates concretely.

20 Conclusion

Concrete-domain participation is not introduced as a primitive ontological assumption. The central result is non-dischargeability: realized repair-localized bounded persistence cannot be preserved as the same fixed-domain target under abstract-only, formal-only, trace-only, semantic-only, skin-only, global, diffuse, AMetric, scope-changing, or bookkeeping closure. Concrete-domain participation is the admissible closure forced when realized persistence, repair-localized bounded standing, and fail-closed continuation are held fixed and shown to be non-dischargeable into abstract-only, formal-only, trace-only, semantic-only, skin-only, global, diffuse, AMetric, scope-changing, or bookkeeping closure.

The theorem therefore yields the concrete-reality conclusion in its fixed-domain AASC form:

$$\text{RealPers}_D \wedge \text{RepairLoc}_D \wedge \text{BoundStand}_D \rightarrow \neg \text{NonConcClosure}_D,$$

with $[\mathcal{C}_D]_{\equiv_{\text{Conc}}}$ as the admissible same-domain closure class. It does not identify the concrete-domain class with a particular universe, physical theory, spacetime model, object inventory, observer, or biological process. It proves that realized repair-localized bounded persistence is structurally non-dischargeable into abstraction: the fixed target cannot remain merely described, merely represented, merely traced, merely referenced, merely formal, globally diffuse, or AMetric without losing the standing-bearing participation that made it realized. Concrete-domain participation is therefore the admissible closure, not an initial ontological posit.

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